



Designing Database using ER Model

Agenda

- ▶ **Database Design**
- ▶ **ER Model Basics**
- ▶ **Relationships in ER Model**
- ▶ **Constraints**
- ▶ **ER Diagram Notations**
- ▶ **Example**

Database Design

Database Design

- ▶ **The task of creating a database application is a complex one, involving:**
 - ▶ Design of the database schema
 - ▶ Design of the programs that access and update the data
 - ▶ Design of a security scheme to control access to data.
- ▶ **The needs of the users play a central role in the design process.**

Database Design

- ▶ **Several phases are required for a complete database design:**
 - ▶ **User Requirements Specification:** Involves interaction with the users.
 - ▶ **Conceptual Design:** User requirements are translated into a conceptual schema of the database.
 - ▶ **Specification of functional requirements:** Users describe the kinds of operations (or transactions) that will be performed on the data.
- ▶ **The output of these phases is a fully specified conceptual design of the database.**

Database Design

- ▶ In the logical-design phase, the designer maps the high-level conceptual schema onto the implementation data model of the database system that will be used.
- ▶ The implementation data model is typically the relational data model.
- ▶ This step typically consists of mapping the conceptual schema defined using the entity-relationship model into a relation schema.

Database Design

- ▶ Finally, the designer uses the resulting system-specific database schema in the subsequent physical-design phase, in which the physical features of the database are specified.
- ▶ These features include the form of file organization and choice of index structures.

Database Design – Requirements

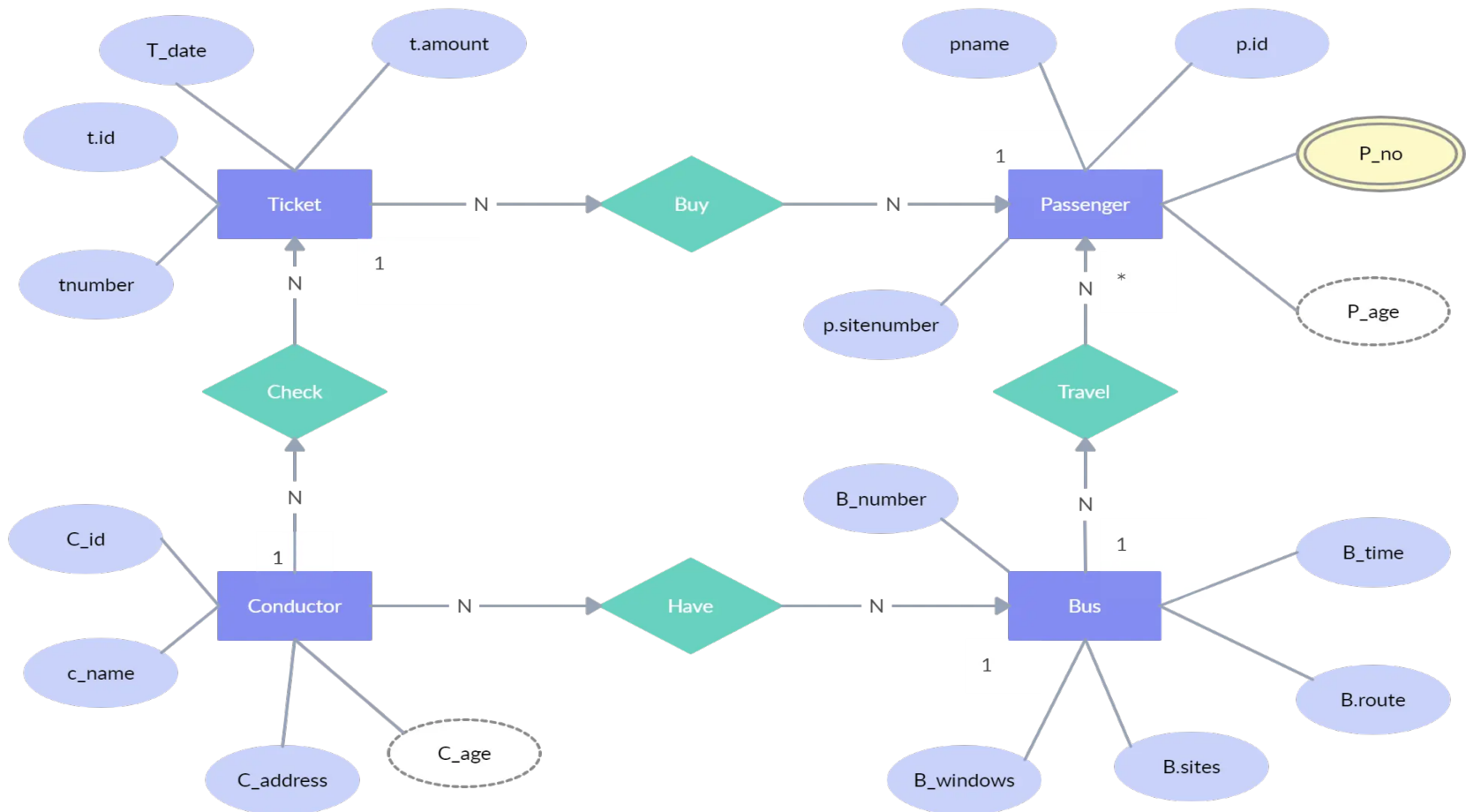
3.2 Project Scope

[Describe what work is in scope for the project, and specifically what work is out of scope... beyond the current budget, resources and timeline as approved by the project stakeholders. This is designed to prevent “scope creep” of additional features and functions not originally anticipated.]

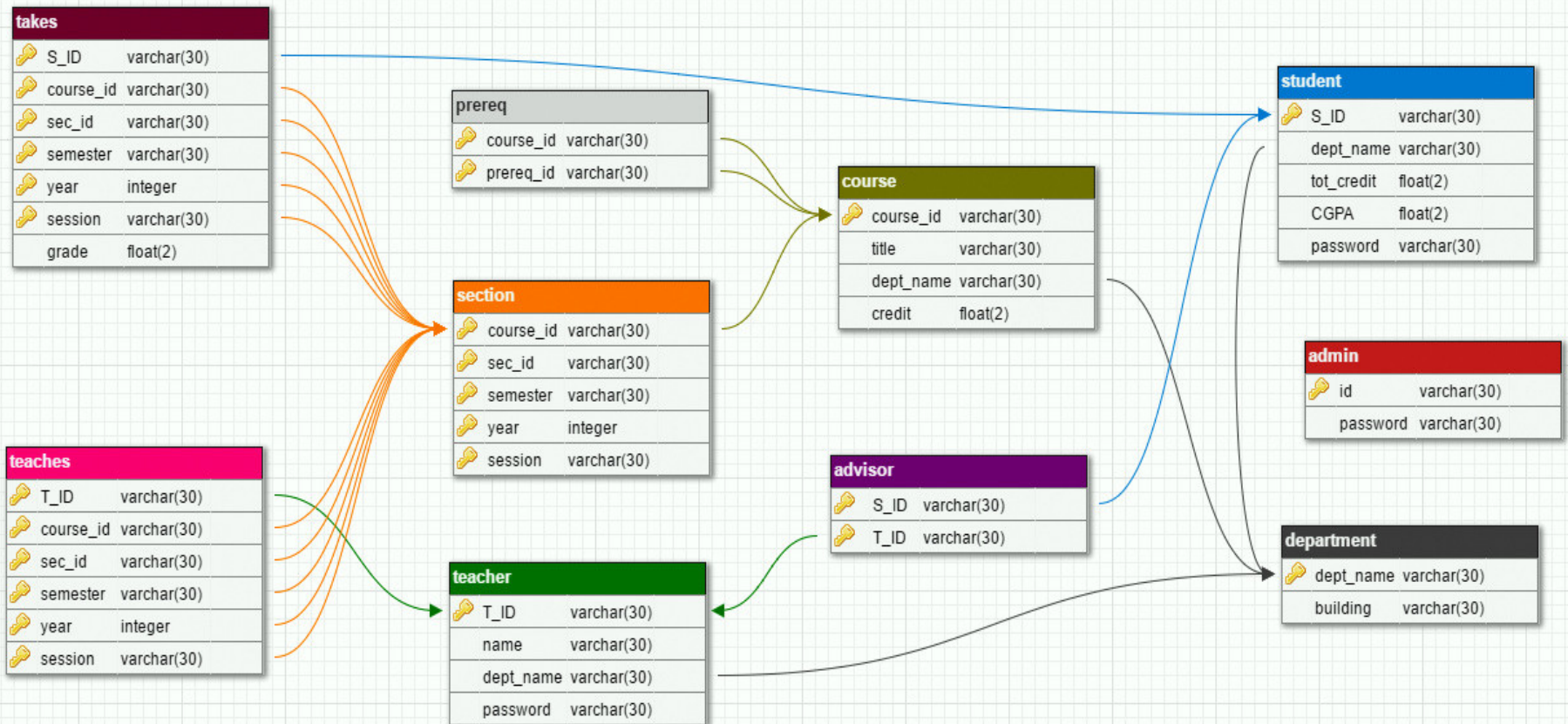
3.2.1 In Scope Functionality

- Create name records for widgets by category
 - Supply Chain
 - Production Lines
 - Internal web apps
 - External web apps
- Ability to create/delete widget names restricted by role
- Search by name, team, date, last modified
- Synchronize widgets across product/operations lines
- Provide audit trail
- Reporting on new, modified, and archived widgets by time period and team

Database Design – ERD



Database Design – Relational Schema



ER Model Basics

Entity-Relationship Model

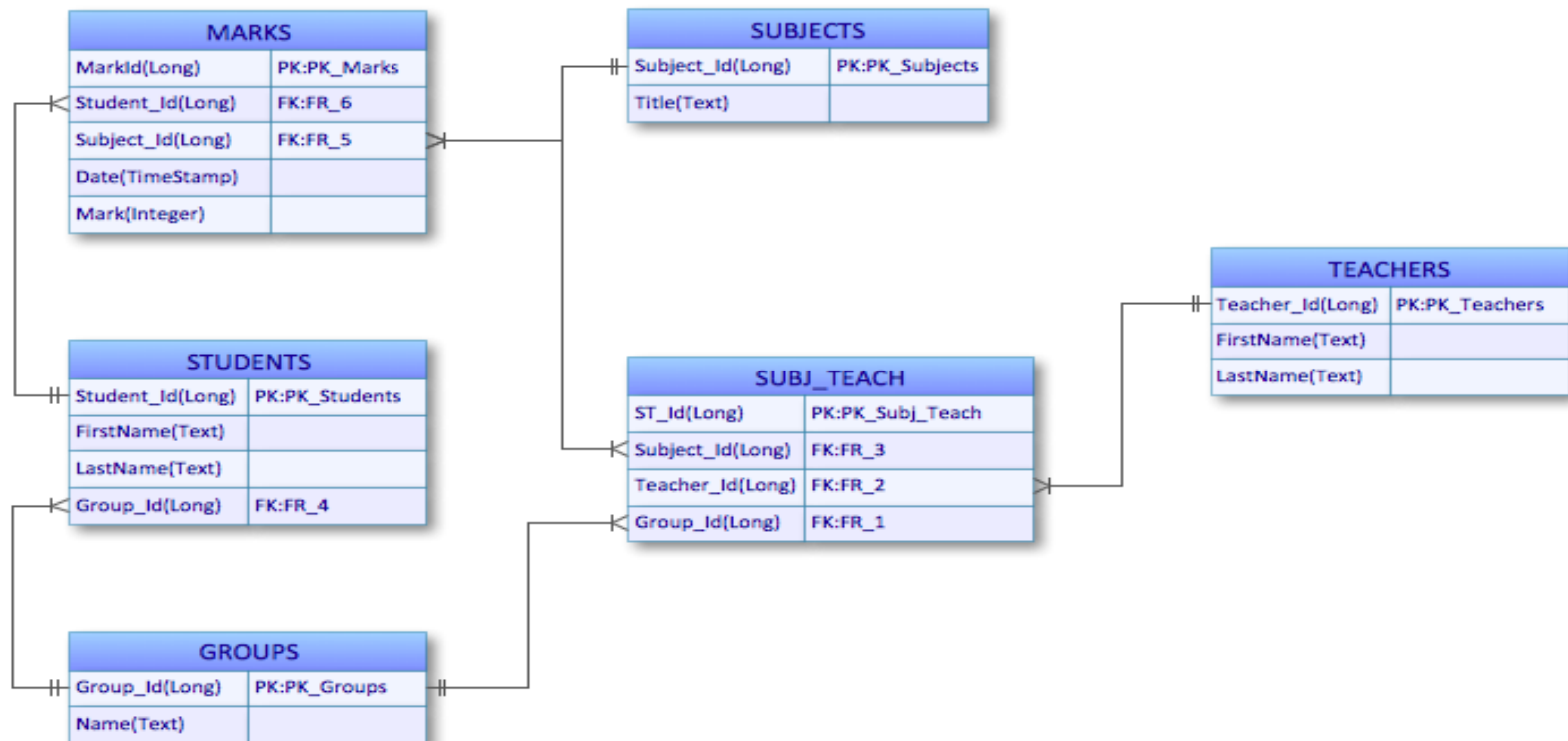
- ▶ The **Entity-Relationship Model (ER Model)** is one of the most widely used **conceptual data models** for designing databases. It was introduced by Peter Chen in 1976.
- ▶ It provides a **high-level, graphical way** of representing the data and its relationships before converting it into tables in a relational database.

Entity-Relationship Model

- ▶ Uses a collection of basic objects called ***entities*** and their ***relationships*** among each other
- ▶ An ***entity*** can represent any thing or object of the real world
- ▶ Depicted by E-R Diagrams

Entity-Relationship Model

Entity-Relationship Diagram using Crow's Foot notation



Entity-Relationship Model

- ▶ This data model is used to represent the overall logical structure of a database
- ▶ Useful in mapping the meanings and interactions of real-world enterprises onto a conceptual schema
 - ▶ Typically includes only the main concepts and the main relationships of the different relations in a database
 - ▶ It gives a first-cut model, with insufficient detail to build an actual database
 - ▶ It hides the internal details of physical storage and targets on describing entities, datatypes, relationships and constraints.

Entity-Relationship Model

- ▶ **The E-R data model employs three basic concepts:**
 - ▶ **Entity Sets**
 - ▶ **Relationship Sets**
 - ▶ **Attributes**
- ▶ **Many database design tools draw on concepts from the E-R model**
- ▶ **The associated diagrammatic representation, commonly used by database designers is the E-R Diagram**

Entity Sets

- ▶ **An entity is a thing or any object of the real world, that is distinguishable from all other objects**
 - ▶ **Example:**
 - ▶ **A person in university – instructor and/or student**
 - ▶ **A course being offered in a semester**
- ▶ **Each entity has one or more properties, i.e. attributes, and values of these attributes are used to uniquely identify a single entity**
 - ▶ **Example:**
 - ▶ **A person has a CNIC number that uniquely identifies him/her from others**

Entity Sets

- ▶ An entity does not only contain these uniquely identified properties but others as well to describe the objects and their transactions completely
- ▶ Hence, it is represented by a set of attributes, which are descriptive properties possessed by all members of an entity set
 - ▶ Instructor = {ID, Name, Department, Salary}
 - ▶ Course = {CourseID, Title, CreditHours, Department}
 - ▶ Book = {BookID, Title, Author, Genre}
- ▶ The subset of these attributes that uniquely identify each entity form the set of primary keys

Entity Sets

- ▶ An entity set is represented in an E-R diagram by a rectangle, which is divided into two parts:
 - ▶ The first part contains the name of the entity set.
 - ▶ The second part contains the names of all the attributes of the entity set.
- ▶ Similar to Schema Diagrams of Relational Model

Entity Sets

- ▶ The E-R diagram below shows two entity sets instructor and student.



Figure 6.1 E-R diagram showing entity sets *instructor* and *student*.

- ▶ The attributes associated with instructor are ID, name, and salary.
- ▶ The attributes associated with student are ID, name, and tot_cred.
- ▶ Attributes that are part of the primary key are underlined

Relationship Sets

- ▶ A relationship is an association among several entities.
- ▶ Example:
 - ▶ A relationship of an instructor with his/her student – advising
 - ▶ A relationship of customer with a bank – account_holder
 - ▶ Relationship(s) between persons and film – acting, directing, etc
 - ▶ A relationship between student and section – takes that course

Relationship Sets

- ▶ A relationship set will then represent a set of relationships of the same type.
 - ▶ Example:
 - ▶ All instructors who are advising, and their students
 - ▶ All the account holders of a specific branch of a bank
 - ▶ All the actors of a film

Relationship Sets

- The following diagram shows a relationship between an instructor and a student

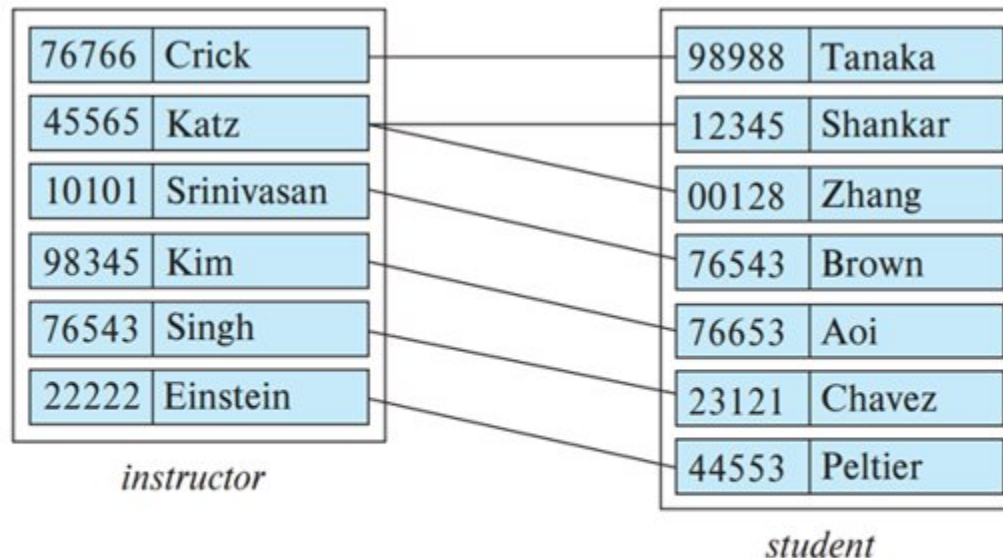


Figure 6.2 Relationship set *advisor* (only some attributes of *instructor* and *student* are shown).

Relationship Sets

- ▶ A relationship instance in an E-R schema represents an association/relationship between the named entities in the real-world enterprise that is being modeled.
- ▶ A relationship set is represented in an E-R diagram by a diamond, which is linked via lines to a number of different entity sets (rectangles).

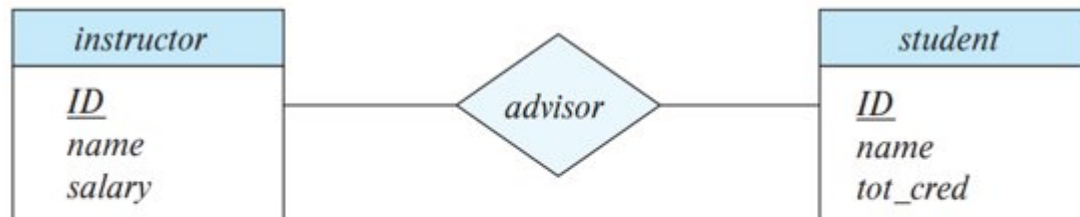


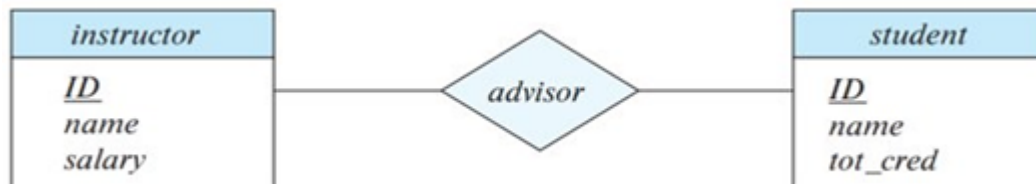
Figure 6.3 E-R diagram showing relationship set *advisor*.

Relationship Sets – Formal Definition

- ▶ Formally, a relationship set is a mathematical relation on $n \geq 2$ (possibly non-distinct) entity sets.
- ▶ If $E_1, E_2, \dots, E_n$ are entity sets, then a relationship set R is a subset of $\{(e_1, e_2, \dots, e_n) \mid e_1 \in E_1, e_2 \in E_2, \dots, e_n \in E_n\}$ where $(e_1, e_2, \dots, e_n)$ is a relationship instance.
- ▶ The association between entity sets is referred to as participation; i.e., the entity sets $E_1, E_2, \dots, E_n$ participate in relationship set R .

Roles

- ▶ The function that an entity plays in a relationship is called that entity's role.
- ▶ Roles are implicit and are not usually specified.
- ▶ However, they are useful when the meaning of a relationship needs clarification.
- ▶ For example, in the instructor and student relationship:



Roles

- ▶ However, when the entity sets of a relationship set are not distinct; i.e., the same entity set participates in a relationship set, in different roles, the roles play an important role
- ▶ In this type of relationship set, sometimes called a recursive relationship set, explicit role names are necessary to specify how an entity participates in a relationship instance.

Roles

► Example:

- A course records information about all the courses offered in the university
- Each course is a prerequisite for another course, which means the same entity, i.e. course is related to another course
- This is then considered a recursive relationship

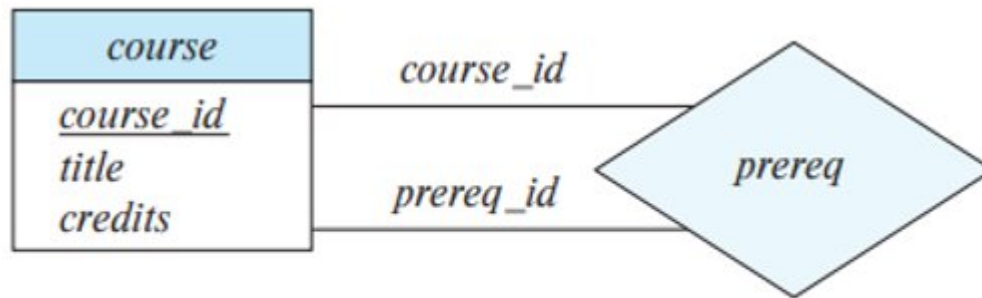


Figure 6.4 E-R diagram with role indicators.

Descriptive Attributes

- ▶ A relationship may also have attributes called descriptive attributes.
- ▶ For example, consider the relationship set takes which relates entity sets student and section
- ▶ We may store a descriptive attribute, grade with the relationship to record the grade that a student received in that section of course offering
- ▶ This depicts that a student gets a certain grade in the section of the course he was taking

Descriptive Attributes

- ▶ An attribute of a relationship set is represented by an undivided rectangle.
- ▶ This rectangle is linked with a dashed line to the diamond representing that relationship set.

Descriptive Attributes

- So for the student-section relationship example from the previous slide, the E-R diagram would be depicted as follows:

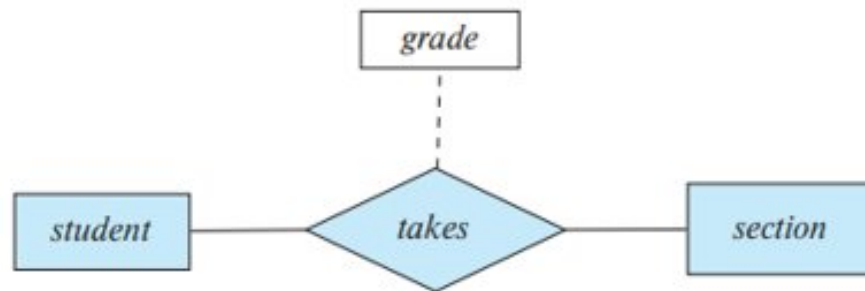


Figure 6.5 E-R diagram with an attribute attached to a relationship set.

- A relationship set may have multiple descriptive attributes

Degree of Relationship

- ▶ Most of the relationship sets in a database system are binary.
- ▶ Occasionally, however, relationship sets involve more than two entity sets.
- ▶ The number of entity sets that participate in a relationship set is the degree of the relationship set.
- ▶ A binary relationship set is of degree 2; a ternary relationship set is of degree 3.

Degree of Relationship

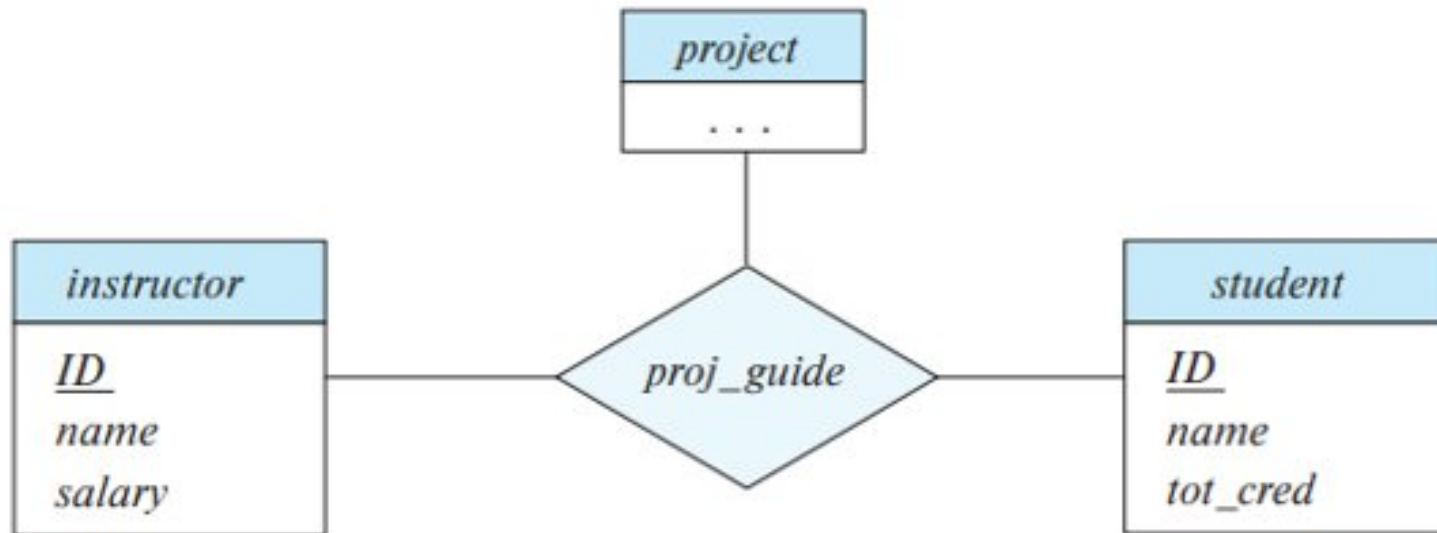


Figure 6.6 E-R diagram with a ternary relationship *proj_guide*.

Attributes

- ▶ An attribute describes a property or characteristic of an entity or a relationship.
- ▶ For each attribute, there is a set of permitted values, called the domain, or value set, of that attribute
- ▶ Each attribute instance must lie within its domain for the database to be correctly defined and become functional
- ▶ For example, the domain of attribute, `course_id` might be the set of all text strings of a certain length.
- ▶ Or, the value set of attribute, `semester` might be strings from the set {Fall, Winter, Spring, Summer}

Attributes

- ▶ As each entity set contains several attributes, then each entity can be described by a set of (attribute, data value) pairs, with one pair for each attribute in the entity
- ▶ For example, a particular instructor entity may be described by the set:
 - ▶ {(ID, 76766), (name, Crick), (dept_name, Biology), (salary, 72000)}
- ▶ This is where the integration of the conceptual schema with the actual enterprise starts being modeled

Attributes

- ▶ An attribute can be characterized into following types:
 - ▶ Simple
 - ▶ Composite
 - ▶ Single-valued
 - ▶ Multivalued
 - ▶ Derived

Attributes - Simple

- ▶ Simple attributes are atomic – they cannot be divided into subparts.
 - ▶ Example:
 - ▶ Instructor ID is a 5-digit integer. It cannot be divided further, hence it's atomic, and thus considered a simple attribute
 - ▶ Department Name is a string, which is also representing a singular value, thus defined as a simple attribute

Attributes - Composite

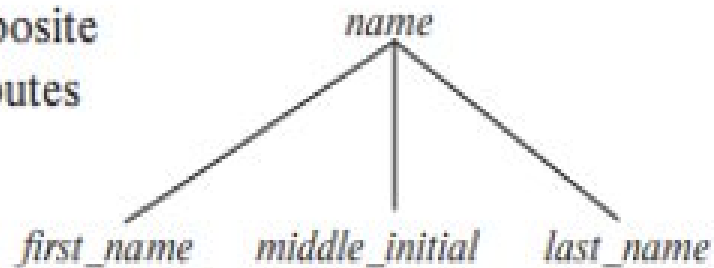
- ▶ Composite attributes can be divided into subparts.
 - ▶ Example:
 - ▶ Instructor Name could be structured as composite attribute, if the name consisted of first name, middle name and last name all together
 - ▶ Phone number can also be defined as a composite attribute, by specifying country code, city code, and/or extension separately
- ▶ Specified as indented values in the E-R diagram

Attributes - Composite

- ▶ A composite attribute may also appear as a hierarchy.
- ▶ The further attributes of each composite attribute are called the component attributes
 - ▶ Example: An address can be a composite attribute, containing street, city, state, and zip code, in which street can become the component attribute that can be further divided into street number, street name and apartment number, forming a hierarchical structure.

Attributes - Composite

composite
attributes



component
attributes

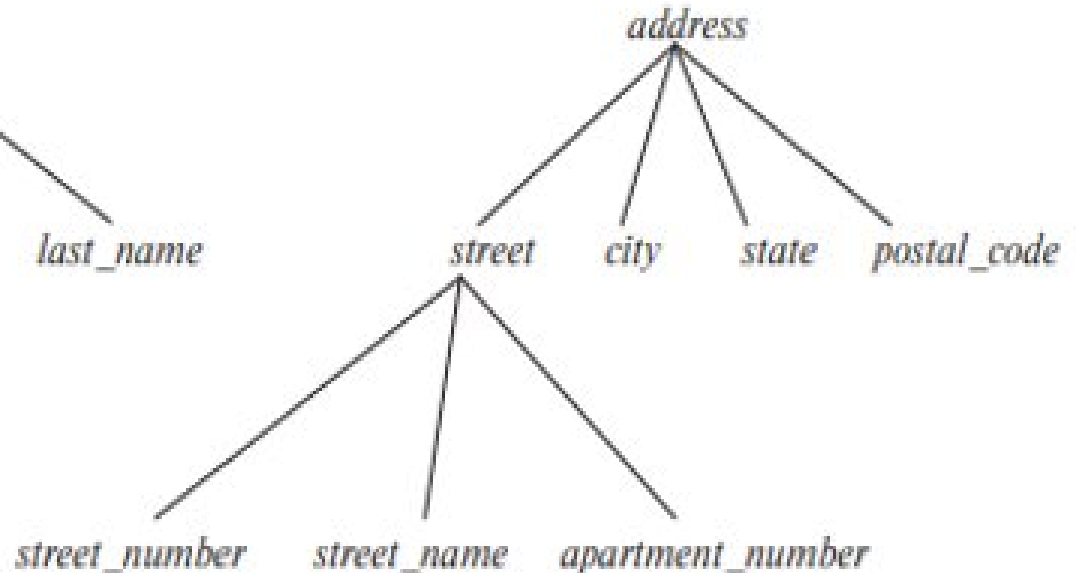
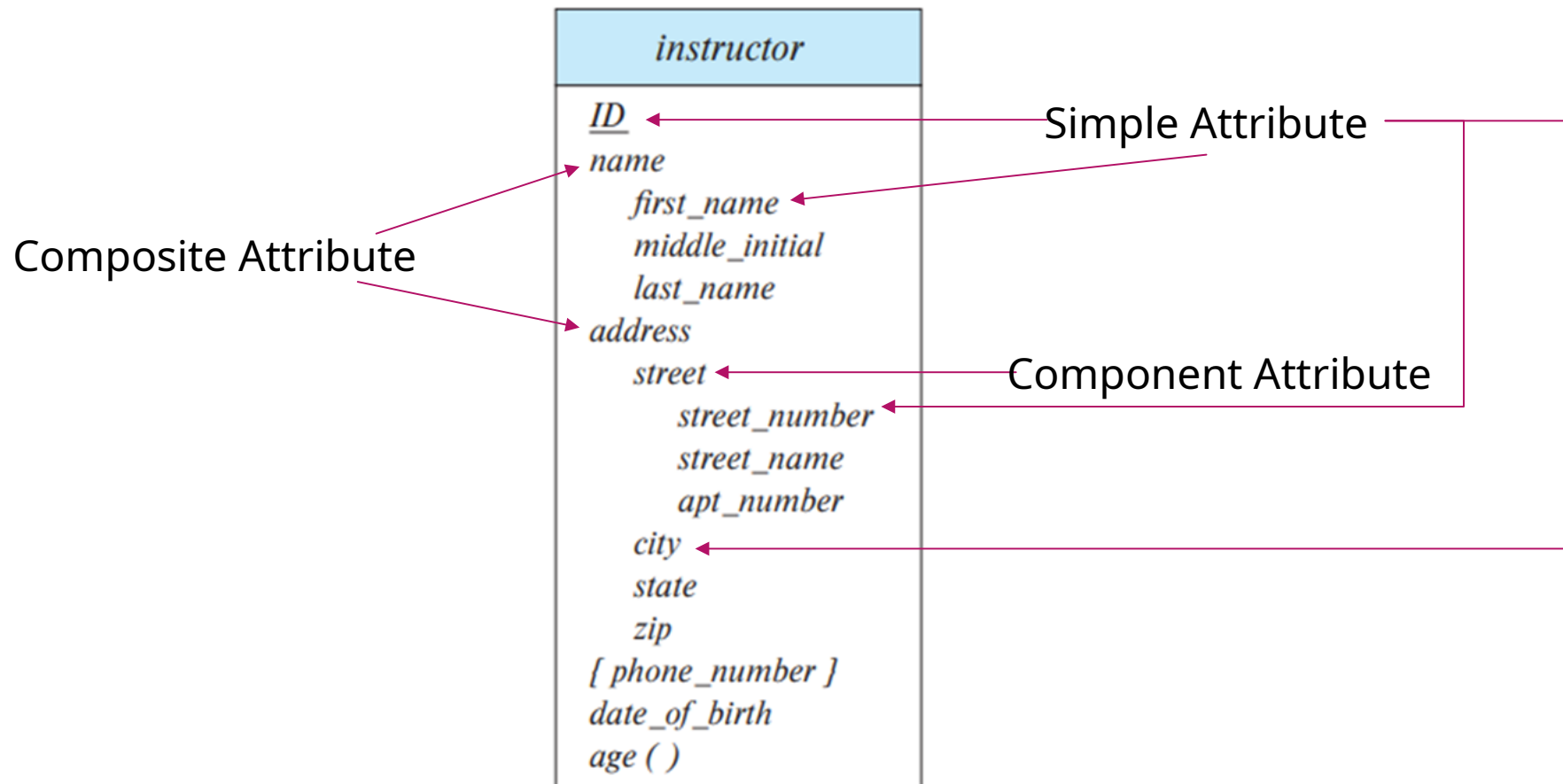


Figure 6.7 Composite attributes instructor *name* and *address*.

Simple and Composite in ER



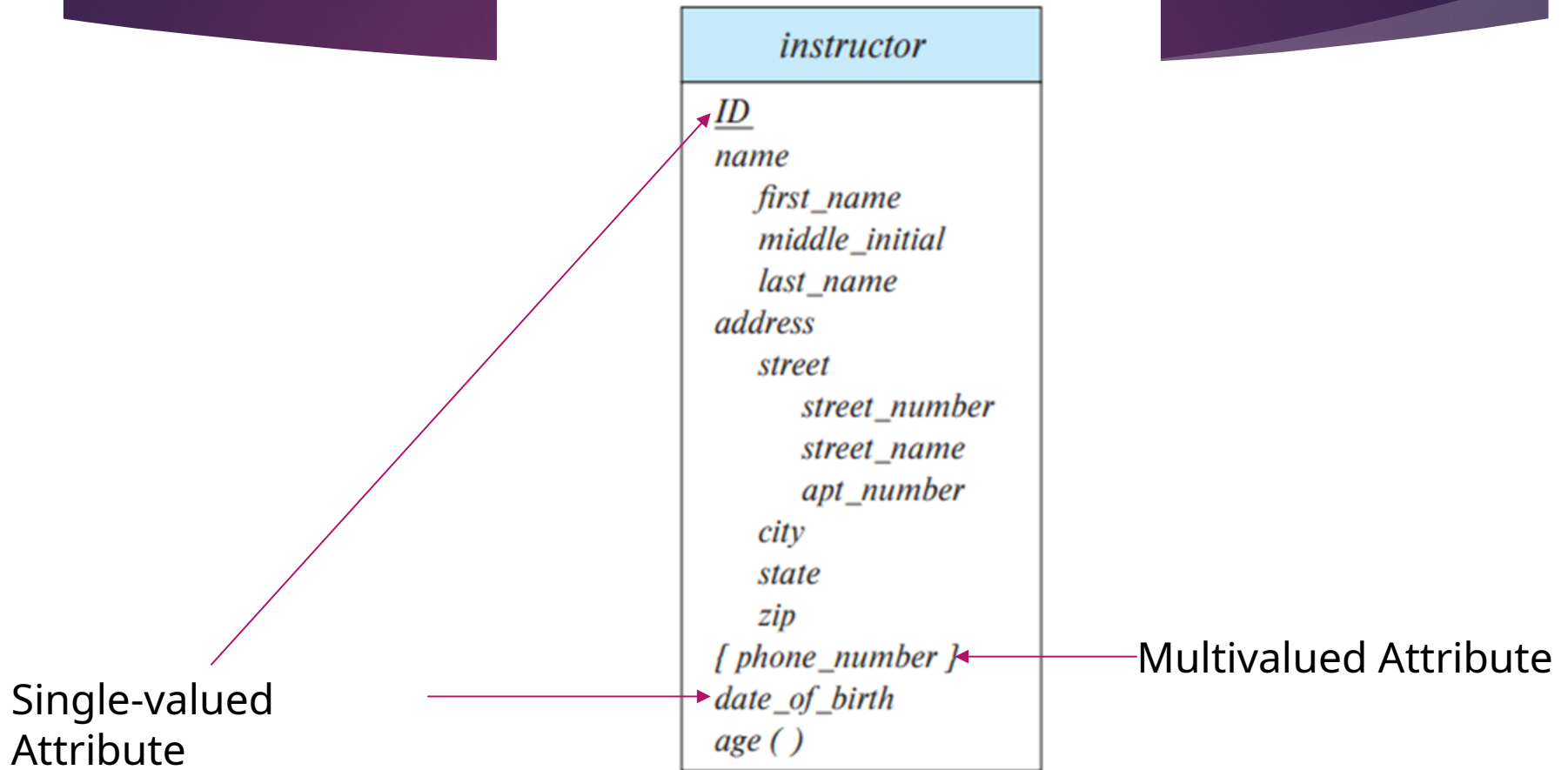
Attributes – Single Valued

- ▶ **Single-valued attributes have a single value for a particular entity. Example:**
 - ▶ **Student ID for a specific student entity refers to only one Student**
 - ▶ **Instructor's Joining Date is also single-valued because there can only be one day when they started, and will remain unchanged**

Attributes – Multi Valued

- ▶ Multivalued attributes, on the other hand, may have zero, one or several values. Example:
 - ▶ Phone number attribute for a person – they can have one or more phone numbers, or they can change their phone numbers
 - ▶ An employee can have multiple dependents for their health insurance policies, including their spouses, children, etc
- ▶ Represented in braces in E-R diagrams. For example, {phone_number}, {dependents}, etc

Single & Mutli Valued in ER



E-R diagram with multivalued attributes.

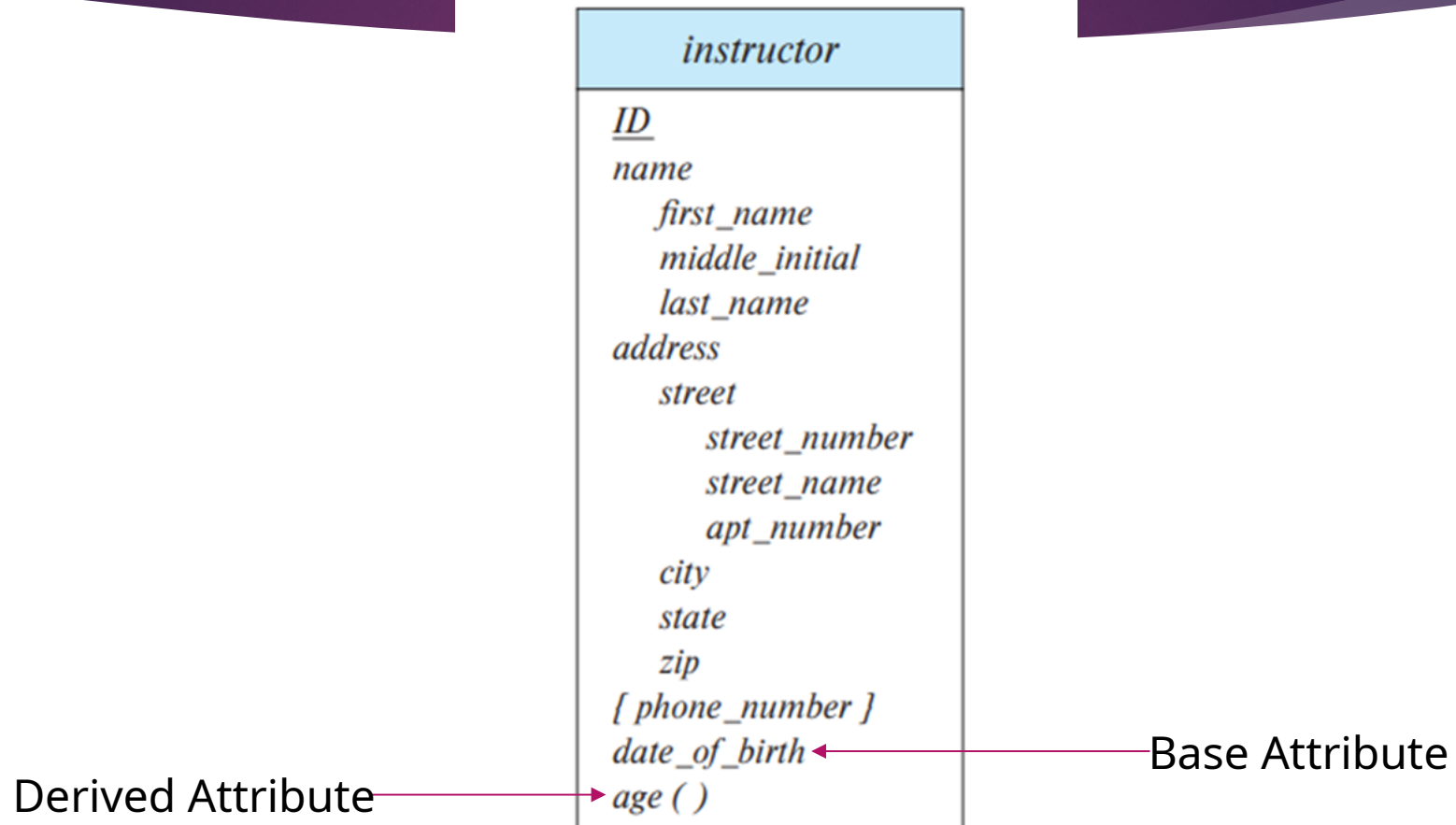
Attributes – Derived

- ▶ An attribute that does not exist in the entity set
- ▶ The value, however, can be derived from the other related attributes.
- ▶ Such attributes are defined as the derived attributes
- ▶ The attribute used for such a computation may be called a base attribute or a stored attribute.
- ▶ The derived attribute may not be stored but computed when needed.
- ▶ Similar to functions in a program

Attributes – Derived

- ▶ **Example:**
- ▶ **Each instructor entity has an attribute which represents the student being advised. We can derive an attribute called `students_advised`, which represents the number of students advised by the instructor. This can be done by counting the number of student entities associated with that particular instructor**
- ▶ **If a student has an attribute, `DOB`, a derived attribute, i.e. `age` can be calculated by comparing that student's date of birth with the current date**

Attributes – Derived



E-R diagram with derived attribute



Relationship in ER Model

Relationship

- ▶ A relationship is an association among several entities.
- ▶ Example:
 - ▶ A relationship of an instructor with his/her student – advising
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 - ▶ Relationship(s) between persons and film – acting, directing, etc
 - ▶ A relationship between student and section – takes that course

Cardinalities

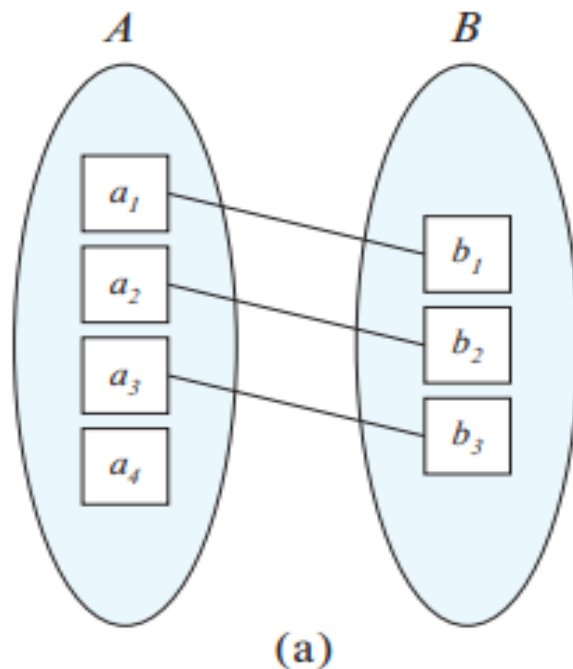
- ▶ Mapping cardinalities, or cardinality ratios, express the number of entities to which another entity can be associated via a relationship set.
- ▶ Mapping cardinalities are most useful in describing binary relationship sets
- ▶ They can also contribute to the description of relationship sets that involve more than two entity sets.

Cardinalities

- ▶ For a binary relationship R between entity sets A and B , the mapping cardinality can be one of the following:
 - ▶ One-to-one
 - ▶ One-to-many
 - ▶ Many-to-many

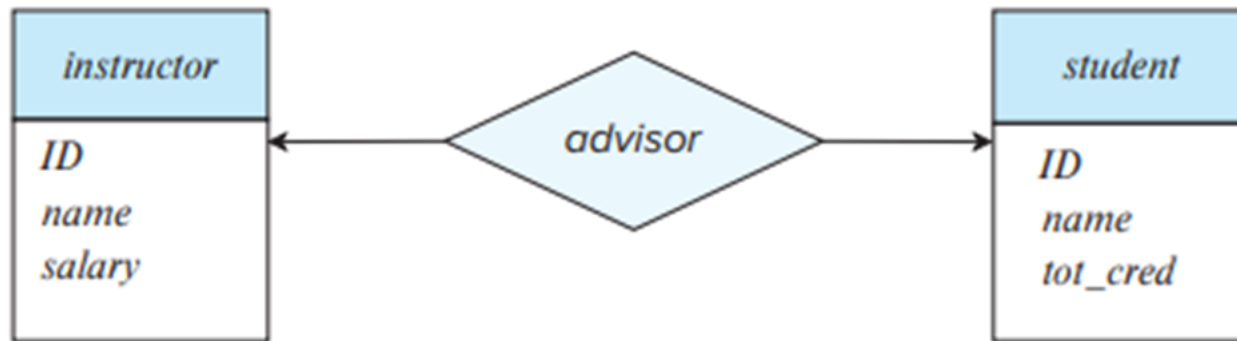
Cardinalities – One-to-One

- An entity in A is associated with at most one entity in B , and an entity in B is associated with at most one entity in A .



Cardinalities – One-to-One

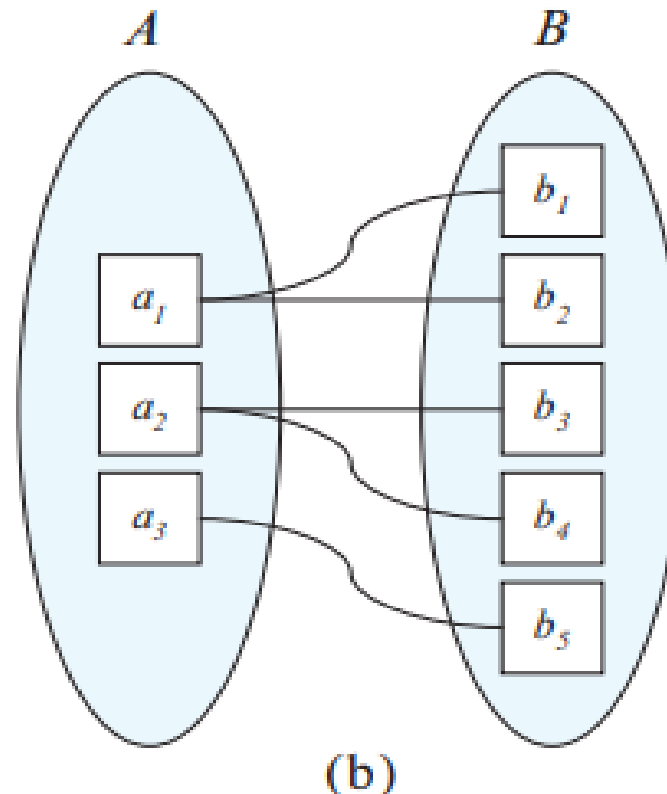
- ▶ **Examples:**
 - ▶ John Doe has a British passport
 - ▶ Every employee is entitled for one company car
- ▶ **ERD Representation of One-to-One cardinality:**
 - ▶ For example, Peter is the faculty advisor of John



(a) One-to-one

Cardinalities – One-to-Many

- An entity in A is associated with any number (zero or more) of entities in B. An entity in B, however, can be associated with at most one entity in A.



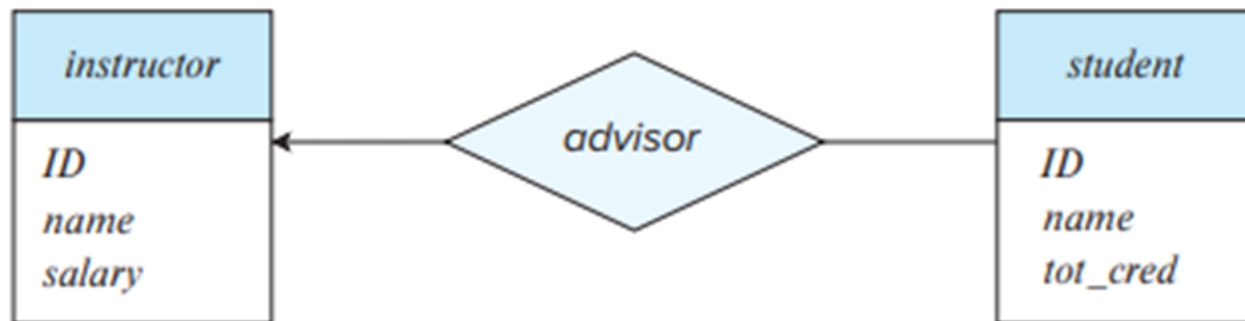
Cardinalities – One-to-Many

► Examples:

- JK Rowling is the author of Harry Potter series
- Jane Smith placed 3 orders on Amazon within a week
- Marketing department has 10 employees

► ERD Representation of One-to-Many cardinality:

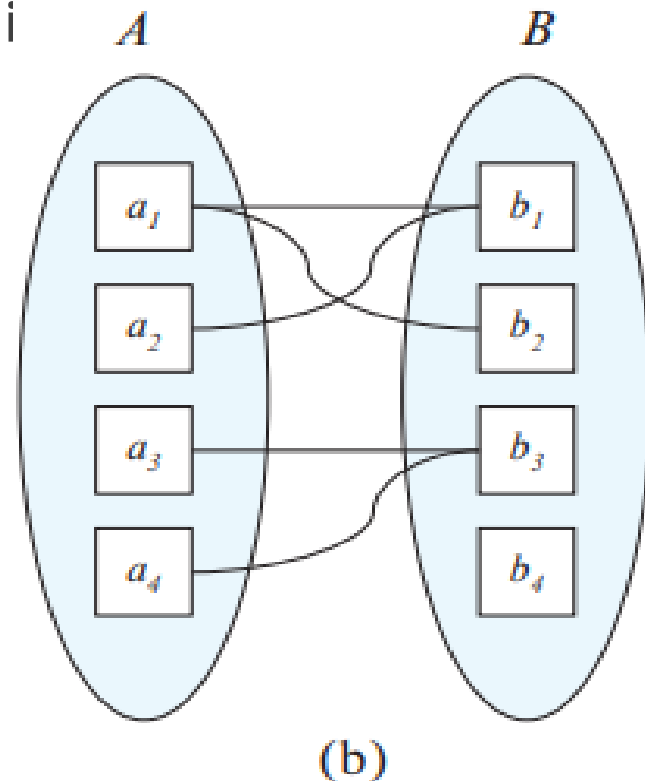
- For example, Peter is assigned 19 advisees this year



(b) One-to-many

Cardinalities – Many-to-Many

- An entity in A is associated with any number (zero or more) of entities in B, and an entity in B is associated with any number (zero or more) of entities in A



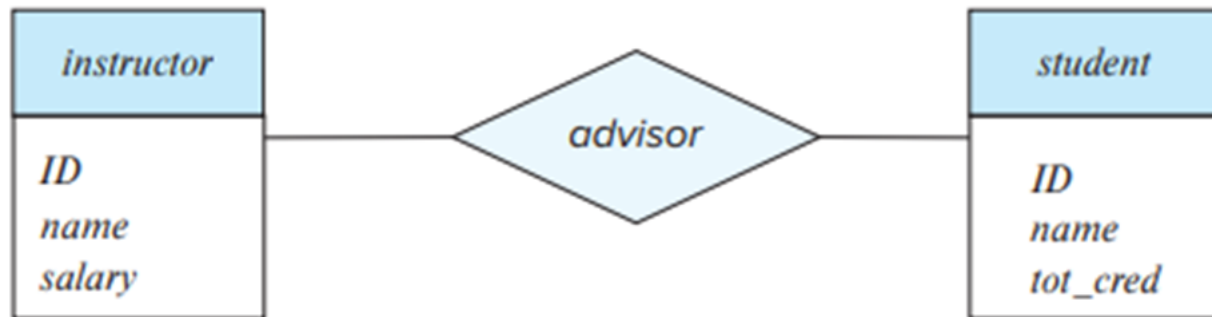
Cardinalities – Many-to-Many

► Examples:

- Each CS Sophomore student is enrolled in Database Systems, Object-Oriented Programming and Digital Logic Design courses
- Best Buy sells different models of laptops

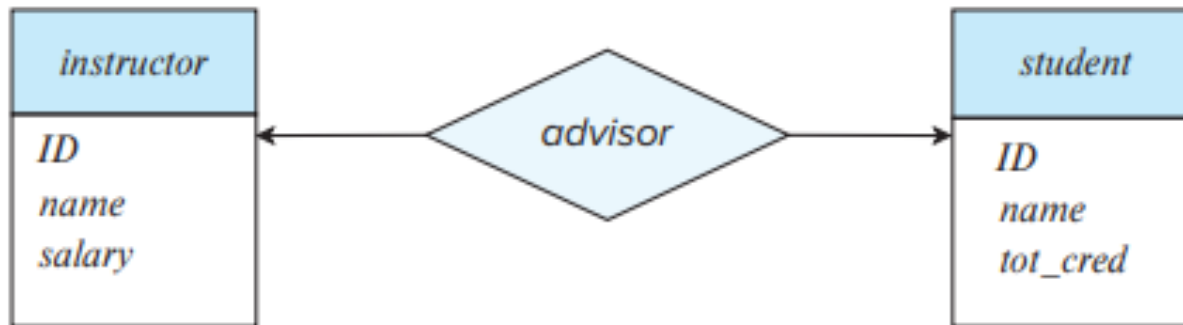
► ERD Representation of Many-to-Many cardinality:

- For example, The CS department instructors are always guiding the students

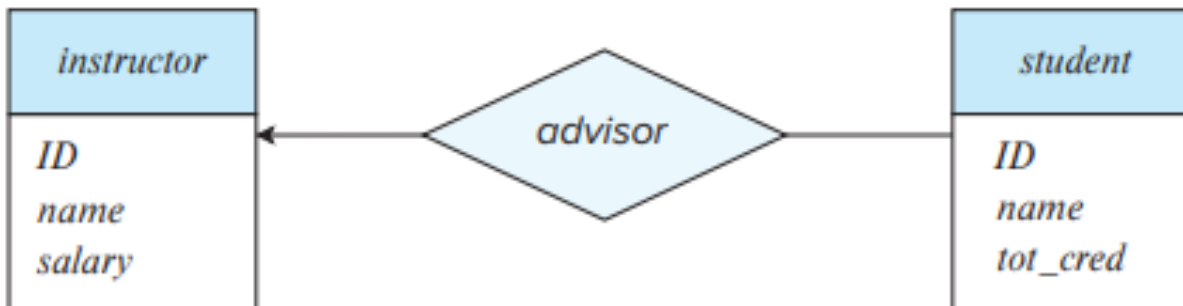


(d) Many-to-many

Cardinalities – ERD Representation

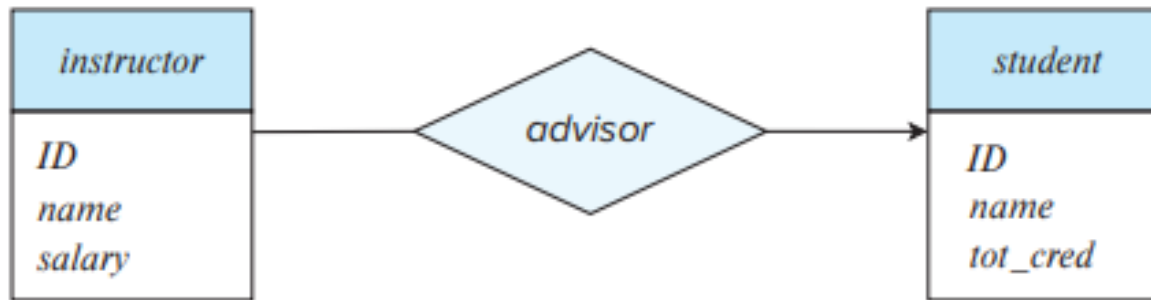


(a) One-to-one

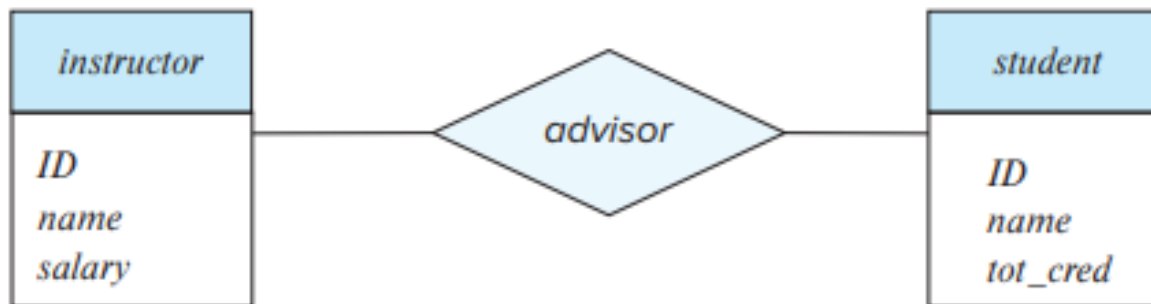


(b) One-to-many

Cardinalities – ERD Representation



(c) Many-to-one



(d) Many-to-many

Total Participation

- The participation of an entity set E in a relationship set R is said to be total if every entity in E must participate in at least one relationship in R. (double lines used to indicate these).

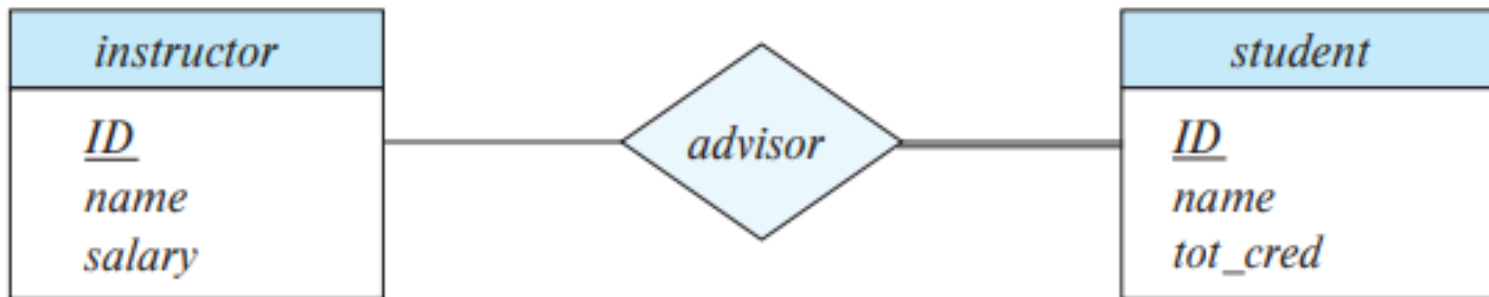
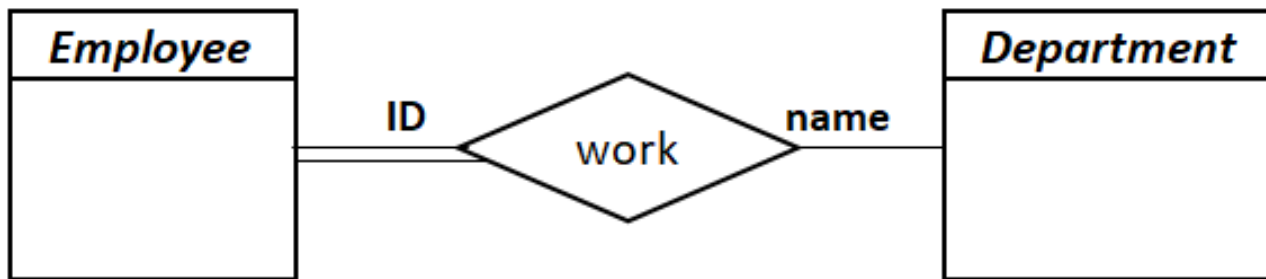


Figure 6.12 E-R diagram showing total participation.

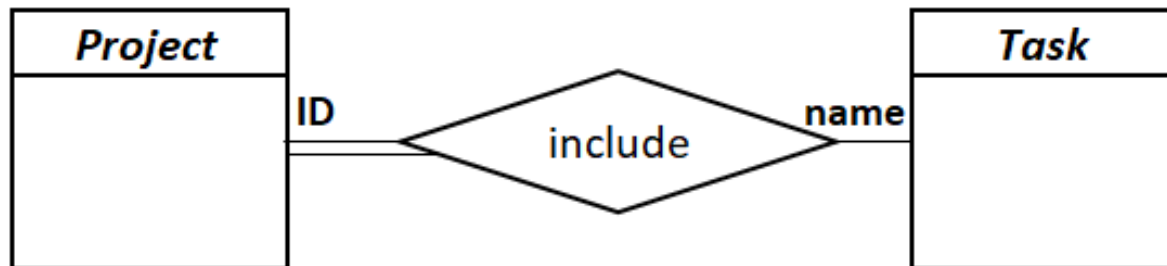
Partial Participation

- If it is possible that some entities in E do not participate in relationships in R, the participation of entity set E in relationship R is said to be partial.



Total & Partial Participation

- ▶ **Examples:**
 - ▶ A project includes tasks
 - ▶ Every project must include at least one task, so Project has total participation in the “includes” relationship
 - ▶ A task does not need to be part of every project, so Task has partial participation in the “includes” relationship





THANK YOU